OSSP – Practical 3

Created the program Practical-3.c using nano, then compiled and ran it. The program uses fork() to create a child process. Both the parent and child print their Process ID (PID) and Parent Process ID (PPID).

```
manu@Manasvi:~/os_lab$ nano Practical-3.c
manu@Manasvi:~/os_lab$ gcc Practical-3.c
manu@Manasvi:~/os_lab$ ./a.out
Parent is running
Parent ID: 4112
Grandparent ID: 338
Child is running
Child ID: 4113
Parent ID: 337
```

Used the **ps** command to list the processes in the current terminal. The child program a.out (PID 4113) is seen running.

```
manu@Manasvi:~/os_lab$ ps
    PID TTY          TIME CMD
    338 pts/0    00:00:00 bash
   4113 pts/0    00:04:49 a.out
   4134 pts/0    00:00:00 ps
manu@Manasvi:~/os_lab$
```

Used **ps u** to view the process state. The child a.out shows the state **R** (Running) and is using nearly 100% CPU because of its loop.

```
manu@Manasvi:~/os_lab$ ps u
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
manu         338  0.0  0.0   6268  5524 pts/0    Ss   16:56   0:00 -bash
manu         414  0.0  0.0   6136  5432 pts/1    Ss+  16:56   0:00 -bash
manu        4113 99.9  0.0   2768  1064 pts/0    R    18:34   5:04 ./a.out
manu        4144  0.0  0.0   7160  4368 pts/0    R+   18:39   0:00 ps u
manu@Manasvi:~/os_lab$
```

Used the **top** command for a live view. The child process (PID 4113) is at the top with state **R** and 100% CPU, while all other processes are in the **S** (Sleeping) state.

```
manu@Manasvi:~/os_lab$ top
top - 18:40:02 up  1:44,  1 user,  load average: 1.00, 0.68, 0.31
Tasks:  27 total,   2 running,  25 sleeping,   0 stopped,   0 zombie
%Cpu(s):  8.3 us,  0.0 sy,  0.0 ni, 91.7 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7776.6 total,   6775.6 free,    568.0 used,    597.8 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7208.6 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
 4113 manu      20   0    2768   1064    968 R 100.0   0.0   5:21.34 a.out
  337 root      20   0    3196   1252   1108 S   0.3   0.0   0:00.42 Relay(338)
    1 root      20   0   24576  14896  10764 S   0.0   0.2   0:01.78 systemd
    2 root      20   0    3180   2212   2072 S   0.0   0.0   0:00.00 init-systemd(Ub
    9 root      20   0    3180   2044   1940 S   0.0   0.0   0:00.00 init
  126 root      20   0    2888   1944   1816 S   0.0   0.0   0:00.04 chronyd-starter
  132 root      20   0    4472   3128   2880 S   0.0   0.0   0:00.03 cron
  135 message+  20   0    9128   5736   4676 S   0.0   0.1   0:00.28 dbus-daemon
  154 root      20   0   44096  29764  14620 S   0.0   0.4   0:00.32 networkd-dispat
  159 root      20   0   18436   9440   8116 S   0.0   0.1   0:00.16 systemd-logind
  176 syslog    20   0  220548   5852   4576 S   0.0   0.1   0:00.13 rsyslogd
  227 root      20   0    5492   2720   2532 S   0.0   0.0   0:00.00 agetty
  255 _chrony   20   0   23812  11288   7144 S   0.0   0.1   0:00.25 chronyd
  260 root      20   0  123460  32636  18024 S   0.0   0.4   0:00.24 unattended-upgr
  270 _chrony   20   0   12096   2404   1760 S   0.0   0.0   0:00.03 chronyd
  336 root      20   0    3196   1120    980 S   0.0   0.0   0:00.00 SessionLeader
  338 manu      20   0    6268   5524   3856 S   0.0   0.1   0:00.37 bash
  339 root      20   0    8644   5592   4752 S   0.0   0.1   0:00.01 login
  382 manu      20   0   21932  10968   8752 S   0.0   0.1   0:00.28 systemd
  384 manu      20   0   22912   4084   2084 S   0.0   0.1   0:00.00 (sd-pam)
  414 manu      20   0    6136   5432   3864 S   0.0   0.1   0:00.02 bash
 1077 polkitd   20   0  306444   8560   7680 S   0.0   0.1   0:00.18 polkitd
 2502 root      19  -1   50512  14448  13240 S   0.0   0.2   0:00.11 systemd-journal
 2584 root      20   0   33776  10272   8564 S   0.0   0.1   0:00.08 systemd-udevd
 2647 systemd+  20   0   22416  13184  10692 S   0.0   0.2   0:00.05 systemd-resolve
 4034 root      20   0 1866032  15384  12560 S   0.0   0.2   0:00.06 wsl-pro-service
 4154 manu      20   0    8172   6068   3968 R   0.0   0.1   0:00.00 top
```

Used **ls /proc/4113** to list the files the kernel maintains about the running process. These files hold live information such as its status, memory maps, and open file descriptors.

```
manu@Manasvi:~/os_lab$ ls /proc/4113
arch_status  cmdline          environ  io                 loginuid   mounts      oom_adj        personality  sessionid     stat     timens_offsets
attr         comm             exe      ksm_merging_pages  map_files  mountstats  oom_score      projid_map   setgroups     statm    timers
auxv         coredump_filter  fd       ksm_stat           maps       net         oom_score_adj  root         smaps         status   timerslack_ns
cgroup       cpuset           fdinfo   latency            mem        ns          pagemap        sched        smaps_rollup  syscall  uid_map
clear_refs   cwd              gid_map  limits             mountinfo  numa_maps   patch_state    schedstat    stack         task     wchan
manu@Manasvi:~/os_lab$
```

Used **cat /proc/4113/status | grep State** to read the process state directly from the kernel. It shows **State: R (running)**, confirming what ps and top displayed.

```
manu@Manasvi:~/os_lab$ cat /proc/4113/status | grep State
State:  R (running)
manu@Manasvi:~/os_lab$
```

Finally used the **kill** command to terminate the child process. Running ps u again shows a.out is no longer listed, confirming the process has ended (Terminated state).

```
manu@Manasvi:~/os_lab$ kill 4113
manu@Manasvi:~/os_lab$ ps u
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
manu         338  0.0  0.0   6268  5524 pts/0    Ss   16:56   0:00 -bash
manu         414  0.0  0.0   6136  5432 pts/1    Ss+  16:56   0:00 -bash
manu        4238  0.0  0.0   7160  4348 pts/0    R+   18:45   0:00 ps u
manu@Manasvi:~/os_lab$
```